

HYDERABAD CITY POLICE

Office of the Commissioner of Police

AI-Based Automated Quality Workbench

Request for Quotation — Technical Specification Document

Document Reference:	HCP/AI-QW/RFQ/2026/01
Version:	1.1
Issue Date:	April 2026
Purpose:	Solicitation of vendor quotations based on technical specifications
Classification:	FOR VENDOR USE ONLY
Prepared By:	O/o The Commissioner of Police, Hyderabad City Police

This document is issued exclusively to empanelled or shortlisted vendors for the purpose of submitting commercial quotations. It contains full technical and functional specifications for the AI-Based Automated Quality Workbench. Vendors are requested to provide item-wise pricing against the Bill of Quantities template in Section 8. Unauthorised reproduction or distribution is strictly prohibited.

1. Introduction & Background

Hyderabad City Police (HCP) proposes to procure and deploy an AI-powered Investigation Quality Workbench to support Investigating Officers (IOs) in improving the quality, consistency, and legal sufficiency of investigations across the complete case lifecycle. This Request for Quotation (RFQ) document provides prospective vendors with the full technical and functional specifications required to prepare responsive commercial bids.

The platform is envisioned as a single-window digital assistant enabling IOs to draft, review, validate, and manage investigation documents — from initial petition and FIR through scene documentation, interrogation reports, forensic processes, and charge sheet preparation. It is designed as a decision-support and productivity enhancement tool, without introducing additional supervisory layers.

1.1 Objectives

- Improve quality and legal sufficiency of investigation documents
- Reduce inconsistencies and trial vulnerabilities in case files
- Standardise investigation practices while retaining case-specific flexibility
- Reduce administrative burden on Investigating Officers
- Enable continuous learning from court judgments and field feedback

1.2 Scope of Procurement

The procurement covers end-to-end supply, development, deployment, integration, training, and post-deployment support for:

- All hardware infrastructure (servers, storage, networking)
- All required software and licenses
- Complete platform development across ten functional modules
- AI model integration and fine-tuning
- Integration with existing police systems (HRMS)
- User training and technical documentation
- Annual Maintenance and Support (AMC)

2. Platform Overview & Positioning

The platform shall function as a Single-Window AI-Assisted Investigation Workbench for Investigating Officers. It is not a supervisory workflow system — it operates as a decision-support tool, drafting assistant, and quality checker.

2.1 User Roles

#	Role	Responsibilities
1	Investigating Officer (Primary User)	Create and manage cases; upload documents; use AI assistance; generate reports; run quality checks; generate communications; digitally sign outputs
2	Station Clerk / Data Entry Operator	Digitize and upload documents; initial data entry; scan and tag exhibits; manage document ingestion queue
3	AI Admin (Critical Role)	Review AI-flagged uncertainties; provide clarifications; train/refine models; maintain checklist libraries; maintain SOP knowledge base; approve new templates; manage staging-to-production promotion
4	System Administrator	Manage user access, security policies, backups, and infrastructure health

No supervisory police role is required in the system workflow. Senior IO or prosecutor review of critical outputs may occur outside the system as per departmental policy.

3. Functional Requirements — Core Platform Modules

The platform shall comprise ten functional modules as detailed below. Vendors must confirm compliance against each requirement in their bid submission.

Module 1 — Case Workbench (Core Hub)

Purpose: Single dashboard per case providing a unified interface for all case management activities.

#	Requirement	Compliance (V/P/N)
1.1	Create case with Crime No. / Petition No. and auto-sync to CCTNS	
1.2	Tag offence type at case creation	
1.3	Attach documents via drag-and-drop, bulk upload, and offline queue	
1.4	Timeline view of all case activities with full version history	
1.5	Action tracker with deadlines and reminders	

Module 2 — Document Quality & Sufficiency Engine

#	Requirement	Compliance (V/P/N)
2.1	IO uploads document and AI evaluates against structured checklists	
2.2	Use Retrieval-Augmented Generation (RAG) to ensure all AI findings are grounded in uploaded document text — no hallucinated facts	
2.3	Output: missing elements with source citations from the document	
2.4	Output: weak areas with actionable improvement suggestions	
2.5	Output: trial-risk indicators with severity classification	
2.6	All suggestions must link to specific document excerpts (citation-grounded)	
2.7	No separate audit report for supervisors — only IO-facing improvement suggestions generated	

Module 3 — Act & Section Recommendation Engine

#	Requirement	Compliance (V/P/N)
3.1	Input: petition / complaint text; output: applicable sections with legal reasoning	
3.2	Identify and list supporting factual ingredients present in the complaint	
3.3	Identify and list missing facts required to sustain recommended sections	
3.4	Provide alternative sections with probability/confidence scoring	
3.5	All recommendations must include disclaimer that final legal determination rests with IO and superior officers	

Module 4 — Document Congruence Engine

#	Requirement	Compliance (V/P/N)
4.1	Cross-document contradiction detection: Petition vs. FIR	
4.2	Cross-document contradiction detection: FIR vs. Witness Statements	
4.3	Cross-document contradiction detection: FIR vs. Charge Sheet	
4.4	Detect timeline inconsistencies across documents	
4.5	Detect accused role mismatches across documents	
4.6	Detect medical report vs. narrative discrepancies	
4.7	Output: conflict alerts with severity levels (High / Medium / Low)	
4.8	Output: missing carry-forward facts between documents	
4.9	IO must be able to dismiss false-positive alerts with reason codes; dismissals feed back into model refinement	

Module 5 — SOP & Plan of Action Generator

#	Requirement	Compliance (V/P/N)
5.1	Auto-detect offence type from case data	
5.2	Generate case-specific investigation steps with legal citations	
5.3	Suggest evidence to collect with forensic requirements	
5.4	Identify and list documents to generate as part of the investigation plan	
5.5	Recommend timelines with statutory deadlines	
5.6	Output: editable Plan of Action with task checklist and completion tracking	
5.7	Output: indicative deadline calendar generation	

Module 6 — Document Generation Engine

The engine must support auto-generation of the following document categories:

Category	Documents
FSL Communications	Forwarding letter to FSL; sample forwarding memo; reminder letters
Evidence Certificates	Section 63 BSA / 65B Certificate; hash value declaration format
Notices	Bank account information request; freeze request; ISP data request; CDR requisition; platform requests (Google/Meta)
Legal Drafts	Arrest memo; seizure memo; remand note; confession recording template

#	Functional Requirement	Compliance (V/P/N)
6.1	Auto-fill: case number, police station, IO details, date, accused details	
6.2	All auto-filled fields must remain editable by the IO	
6.3	Export in both Word (.docx) and PDF formats	

#	Functional Requirement	Compliance (V/P/N)
6.4	Support digital signature / DSC integration on generated documents	

Module 7 — Multilingual OCR & Translation Engine

#	Requirement	Compliance (V/P/N)
7.1	Process handwritten and scanned input documents	
7.2	Language support: Telugu, Urdu, Hindi, English (minimum)	
7.3	Output: extracted text, translated English version, side-by-side view	
7.4	Confidence tagging: High / Medium (review recommended) / Low (mandatory manual review flag)	
7.5	Progressive Web App (PWA) architecture enabling document upload and OCR queuing in low-connectivity/offline environments	
7.6	Automatic sync and processing when connectivity is restored	

Module 8 — AI Training & Feedback Loop

#	Requirement	Compliance (V/P/N)
8.1	AI must flag uncertain outputs with tags such as: 'Ambiguous statement', 'Unclear handwriting', 'Possible legal mismatch', 'Missing ingredient'	
8.2	Dedicated AI Admin queue management dashboard for reviewing flagged cases	
8.3	AI Admin can provide corrections with rationale; updates knowledge base	
8.4	All checklist and SOP updates deployed to staging environment first	
8.5	AI Admin validates staging changes before promotion to production	
8.6	Version tracking for all knowledge base and model updates	
8.7	Full rollback capability for any production update	

Module 9 — Judgment Analysis Engine

#	Requirement	Compliance (V/P/N)
9.1	Upload court judgment (PDF/text) for analysis	
9.2	AI extracts: case facts, issues raised, reasons for acquittal/conviction	
9.3	AI extracts: investigation lapses with specific court observations	
9.4	AI extracts: court observations on evidence quality	
9.5	Output: 'Investigation Lessons' and 'Avoidable Errors' summaries	
9.6	Propose checklist updates (with AI Admin approval before production)	
9.7	Findings feed into checklist engine, SOP module, and training loop	

Module 10 — Usage & Adoption Analytics

#	Requirement	Compliance (V/P/N)
10.1	Track: cases created per IO; documents uploaded; AI checks performed; drafts generated	
10.2	Track: time spent per case; feature usage frequency	
10.3	Generate: officer usage report; feature adoption report; monthly trends	

Compliance column legend: V = Full Compliance | P = Partial Compliance (deviation to be described) | N = Non-Compliant. Vendors must provide written explanation for any P or N responses.

4. Key AI Design Principles (Mandatory)

All AI components of the platform shall adhere to the following non-negotiable design principles. Vendor bids must demonstrate how each principle is achieved.

Principle	Description / Requirement	Vendor Confirmation
No Hallucination Rule	AI must not invent facts. All outputs grounded in source documents using RAG architecture. Each AI finding must include citation to specific text excerpt.	
Evidence-Based Output	Every finding must map to input text with reference — no unsourced statements.	
Suggestive, Not Prescriptive	AI suggests; IO decides. No automatic acceptance of AI outputs permitted.	
Confidence Tagging	All uncertain outputs must be flagged with confidence scores (High / Medium / Low).	
Editable Outputs	All AI-generated drafts and suggestions must be editable by the IO.	
Human-in-the-Loop	High-stakes outputs (section recommendations, evidence certificates) require IO review before finalisation, with an optional flag for senior review per departmental policy (offline or online).	

5. Technical & Infrastructure Requirements

5.1 Deployment Architecture

The platform shall be deployed as a secure, on-premise, AI-enabled application. The architecture must be modular, scalable, and designed to support approximately 100 concurrent users. Integration with HRMS and CCTNS is required.

Parameter	Requirement
Deployment Model	On-premise, containerised microservices (Kubernetes)
Architecture Layers	Application Layer (UI + APIs); Data & Storage Layer; AI & Processing Layer
High Availability	Active-passive cluster configuration with automated failover
RTO (Recovery Time Objective)	≤ 1 hour

Parameter	Requirement
RPO (Recovery Point Objective)	≤ 15 minutes
Backup Strategy	Continuous incremental backups; daily full backups; immutable backups
DR Testing	Quarterly failover drills
Concurrent Users (Design Target)	~100 concurrent users (scalable)

5.2 Hardware Infrastructure Specifications

Vendors must quote against the following minimum hardware specifications. Equivalent or superior alternatives are acceptable with full technical justification.

5.2.1 Application & Database Servers (Active-Passive Cluster)

Component	Minimum Specification	Vendor Proposed Spec
CPU	16–24 cores (Intel Xeon Gold / AMD EPYC class)	
RAM	128 GB DDR5	
Storage (OS)	2 × 1.92 TB NVMe SSD	
Storage (DB)	2 × 3.84 TB NVMe SSD	
Network	10 GbE NIC	
Power	Redundant PSU	
Quantity	2 units (active-passive cluster)	

5.2.2 AI Processing Server

Component	Minimum Specification	Vendor Proposed Spec
CPU	16–24 cores (Intel Xeon Gold / AMD EPYC class)	
RAM	128 GB DDR5	
GPU	1 × NVIDIA RTX 6000 Ada (48 GB VRAM, ECC) or equivalent	
Storage	2 × 1.92 TB NVMe SSD	
Network	10 GbE NIC	
Power	Redundant PSU	
Quantity	1 unit	

5.2.3 Storage & Backup

Component	Minimum Specification	Vendor Proposed Spec
Usable Capacity	12–20 TB (scalable)	
Storage Type	S3-compatible object storage (MinIO) or NAS	
Features	Snapshotting, versioning, immutable backups, archival support	

Component	Minimum Specification	Vendor Proposed Spec
Quantity	1 unit	

5.2.4 Networking & Security Hardware

Component	Minimum Specification	Vendor Proposed Spec
Core Switch	24-port 10 GbE managed switch with PoE+, qty: 2	
Firewall / UTM	Next-gen firewall with IPS, SSL inspection, ≥ 1 Gbps throughput	
Load Balancer	Software (HAProxy/NGINX) or hardware appliance	
UPS	10 kVA online UPS with ≥ 30-minute runtime	
Server Rack	42U server rack with PDU, KVM, cabling	

5.3 Software Stack Requirements

Layer	Component	Requirement / Reference Stack	Vendor Proposed
Frontend	Framework	React (TypeScript) with PWA capabilities	
Frontend	UI Components	Material UI / Ant Design	
Frontend	Offline Support	Service workers for offline queuing; IndexedDB local storage	
Backend	Framework	Python (FastAPI) for API orchestration and business logic	
Backend	Background Jobs	Celery/Redis or Temporal for async processing	
Containerisation	Runtime	Docker	
Orchestration	Platform	Kubernetes with active-passive failover	
API Gateway	Ingress	NGINX Ingress / Traefik	
Database	RDBMS	PostgreSQL v17+ with logical replication for HA	
Object Storage	Storage	MinIO (S3-compatible) with encryption enabled	
Search & Indexing	Engine	OpenSearch / Elasticsearch	
Vector Storage	AI Retrieval	PostgreSQL with pgvector extension	

5.4 Security Requirements

Layer	Standard / Requirement	Compliance (V/P/N)
Data at Rest	AES-256 encryption for databases, object storage, and backups	
Data in Transit	TLS 1.3 for all internal and external communications	

Layer	Standard / Requirement	Compliance (V/P/N)
Key Management	Hardware Security Module (HSM) or equivalent for key storage and rotation	
Access Control	Role-based access control (RBAC) with attribute-based policies for sensitive data	
Audit Logging	Tamper-evident, immutable logs for all user and AI actions (uploads, edits, processing, report generation)	
Document Integrity	SHA-256 hash stored for all uploaded and generated documents	
AI Output Traceability	Each AI-generated suggestion linked to source document excerpt and model version	

5.5 System Integration Requirements

The platform must integrate with the following existing police systems. Vendors must demonstrate prior experience with equivalent integrations and provide their proposed technical approach in the Technical Bid.

System	Integration Method	Data Points Required
HRMS	REST APIs / LDAP	Employee ID, name, rank, posting, authentication

6. Multi-Engine AI Architecture Requirements

6.1 Architecture Principles

The platform shall adopt a multi-engine AI architecture. All AI models handling sensitive case data must be self-hosted on-premise. API-based models may only be used where self-hosted performance is demonstrably insufficient, and only with written approval. The decision on which model and how many models to use will depend on initial procurement cost and annual recurring cost of the solution.

6.2 Task-to-Model Mapping Requirements

Task	Reference/Preferred Engine	Self-Hosted Required	Vendor Proposed Engine	Compliance
OCR (Printed Text)	Tesseract / EasyOCR	Yes		
OCR (Handwritten Text)	Fine-tuned TrOCR / Donut (primary); Gemini API (fallback only)	Partial		
Translation (Indic Languages)	IndicTrans2 / NLLB-200	Yes		
Legal Section Recommendation	Fine-tuned Llama 3 / Mistral (7B–70B)	Yes		
Ingredient Mapping	Fine-tuned Llama 3 / Mistral (7B–13B)	Yes		
Checklist Evaluation	Rule-based engine + Lightweight LLM	Yes		
Congruence Detection	Fine-tuned Llama 3 / Mistral (13B–70B)	Yes		
SOP Generation	Fine-tuned Llama 3 / Mistral (7B–13B)	Yes		
Document Drafting	Template engine + Lightweight LLM	Yes		
Judgment Analysis	Fine-tuned Llama 3 / Mistral (13B–70B)	Yes		

Task	Reference/Preferred Engine	Self-Hosted Required	Vendor Proposed Engine	Compliance
Entity Extraction	SpaCy + Custom NER Model	Yes		
Semantic Search	BGE / E5 Embedding Models	Yes		

6.3 Self-Hosting Infrastructure Requirements

Model Type	GPU Memory Required	Deployment Method
Small Models (7B–8B)	16–24 GB	vLLM / TGI for inference optimisation
Medium Models (13B–20B)	24–48 GB	vLLM with quantisation (4-bit/8-bit)
Large Models (70B+)	48–80 GB	Multiple GPUs or quantisation
Embedding Models	4–8 GB	ONNX Runtime / HuggingFace Inference
OCR Models	4–8 GB	Native libraries
Translation Models	8–16 GB	Fairseq / HuggingFace

6.4 Model Fine-Tuning Requirements

Vendors must include domain-specific fine-tuning of the following models as part of the deployment scope:

Model	Fine-Tuning Dataset	Purpose
Llama 3 / Mistral (Legal)	IPC/BNS sections, ingredient lists, court judgments, investigation SOPs	Section recommendation, ingredient mapping
TrOCR (Handwriting)	Anonymised police handwriting samples (Telugu, Urdu, Hindi, English)	Handwritten document OCR
NER Model (SpaCy)	Anonymised FIRs, charge sheets	Entity extraction (names, dates, locations, case numbers)
Embedding Model	Police documents, legal texts	Semantic search, document similarity

6.5 Performance Targets (AI Services)

Metric	Description	Target	Vendor Committed SLA
Latency p95 (OCR)	Time to complete OCR inference	< 5 seconds	
Latency p95 (LLM Tasks)	Time to complete LLM inference	< 10 seconds	
System Error Rate	Failed inference requests	< 1%	
Fallback Rate	Requests routed to fallback model	< 5%	
GPU Utilisation (Sustained)	Sustained GPU resource usage	< 80%	
Platform Uptime	Overall system availability	≥ 99.5%	

7. AI Training & Prompt Testing Phase

7.1 Overview

Before go-live, all AI prompts governing the system's legal intelligence, checklist evaluation, congruence detection, and document generation capabilities shall undergo a structured Prompt Testing and Firming Up process. This phase must be included in the vendor's bid as a distinct deliverable.

7.2 Prompt Categories to be Tested

Prompt Category	Purpose	Example Use Case
Petition Sufficiency Check	Evaluate if petition has minimum information sufficiency to proceed with police action	Flag petitions missing 5WH or complainant details
Section Recommendation	Analyse complaint text and recommend applicable penal sections	Identifying 498A IPC ingredients from dowry harassment complaint
Ingredient Mapping	Extract factual elements required to sustain a section	'Presence of demand for dowry', 'cruelty', 'harassment'
Checklist Evaluation	Assess document completeness against investigation standards	FIR missing mandatory fields; charge sheet missing witness statements
Congruence Detection	Identify contradictions across documents	FIR timeline vs. witness statement timeline
SOP Generation	Generate case-specific investigation plans	Steps for cybercrime investigation vs. homicide
Judgment Analysis	Extract lessons from court judgments	Reasons for acquittal; investigation lapses cited by court
Document Drafting	Populate templates with case data	FSL forwarding letter; 65B certificate
OCR Post-Processing	Structure and clean extracted text	Organise handwritten FIR into structured fields

7.3 Testing Phases & Timeline

Phase	Description	Duration
Phase 1: Prompt Engineering	Initial prompt design; baseline output generation	2 weeks
Phase 2: Dry Run Testing	Run prompts against test dataset; collect outputs	1 week
Phase 3: Expert Review	Domain experts evaluate outputs for accuracy, relevance, legal correctness	2 weeks
Phase 4: Refinement	Iterative prompt tuning based on expert feedback	2 weeks
Phase 5: Validation	Final run on held-out test set; sign-off	1 week

7.4 Acceptance Metrics for Prompt Sign-Off

Metric	Definition	Target
Accuracy	Output matches expected result per expert review	≥ 90%
Precision	Relevant outputs vs. total outputs	≥ 85%
Recall	Relevant outputs captured vs. total possible	≥ 85%
Hallucination Rate	Outputs containing facts not present in source document	0%

Metric	Definition	Target
Legal Correctness	Outputs comply with applicable laws and procedures	100%
Confidence Calibration	AI confidence scores correlate with actual accuracy	≥ 80% correlation

7.5 Acceptance Criteria for AI Training Phase Completion

Acceptance Criterion	Status
All prompt categories have test datasets with ≥ 200 cases	
All prompts achieve target accuracy metrics	
Zero hallucination observed in all validation tests	
Legal review sign-off obtained for all prompts	
Domain expert (Senior IO) sign-off obtained	
Prompt library documented with full version control	
Annotation interface operational	
AI Admin training completed	
Staging environment validated	
Production deployment readiness approved	

8. Bill of Quantities (BOQ) — Vendor Quotation Template

Vendors are requested to provide firm pricing against each line item below. All prices must be quoted in Indian Rupees (INR) exclusive of GST. GST must be quoted separately at the applicable rate. Blank cells shaded in grey are for vendor completion. Where third party hardware/ software is required, appropriate pricing from publicly available sources may be quoted.

Instructions: (1) Complete all grey-shaded fields. (2) Quote unit prices and total prices for all items. (3) Items marked 'Optional' should be quoted separately and will not be evaluated as part of the base bid. (4) Prices must remain valid for a minimum of 90 days from submission date. (5) Submit the completed BOQ as a separate commercial bid document.

8.1 Hardware Infrastructure

A. Application & Database Servers

S.No	Item	Specification	Qty	Unit Price (INR)	Total (INR)
1	Rack Server (Primary — Active Node)	Intel Xeon Gold 6426Y (16-core) or AMD EPYC 9254 (24-core), 128 GB DDR5 RAM, 2×1.92 TB NVMe SSD (OS), 2×3.84 TB NVMe SSD (DB), 10 GbE NIC, Redundant PSU	1		
2	Rack Server (Secondary — Passive Node)	Same configuration as primary for active-passive failover	1		
3	Kubernetes Worker Nodes	Intel Xeon Silver 4314 (16-core), 64 GB RAM, 2×1.92 TB NVMe SSD, 10 GbE NIC	2		

S.No	Item	Specification	Qty	Unit Price (INR)	Total (INR)
4	Server Rack & Accessories	42U server rack, PDU, KVM, cabling	1		

B. AI Processing Server

S.No	Item	Specification	Qty	Unit Price (INR)	Total (INR)
5	AI Processing Server	Intel Xeon Gold 6426Y (16-core) or AMD EPYC 9254, 128 GB DDR5 RAM, 1×NVIDIA RTX 6000 Ada (48 GB VRAM) or 2×NVIDIA RTX 5000 Ada, 2×1.92 TB NVMe SSD, 10 GbE NIC, Redundant PSU	1		
6	GPU Upgrade (Optional)	Additional NVIDIA RTX 6000 Ada for larger models	1		

C. Storage & Backup

S.No	Item	Specification	Qty	Unit Price (INR)	Total (INR)
7	NAS / SAN Storage	24 TB usable capacity (RAID-6), 10 GbE iSCSI/NFS, dual controller, redundant PSU	1		
8	Backup Appliance	Integrated backup with 12 TB usable, immutable backups, replication support	1		
9	LTO Tape Drive (Optional)	Offline archival for legal compliance	1		

D. Networking & Security

S.No	Item	Specification	Qty	Unit Price (INR)	Total (INR)
10	Core Switch	24-port 10 GbE managed switch with PoE+	2		
11	Firewall / UTM Appliance	Next-gen firewall with IPS, SSL inspection, ≥ 1 Gbps throughput	1		
12	Load Balancer	HAProxy / NGINX or hardware appliance	1		
13	UPS	10 kVA online UPS with ≥ 30-minute runtime	1		

HARDWARE INFRASTRUCTURE TOTAL (Items 1–13, excl. Optional)

8.2 Software & Licenses

S.No	Item	Specification	Qty	Annual Price (INR)	3-Year Total (INR)
14	OS — Red Hat Enterprise Linux	RHEL Server subscription (2 sockets)	5		
15	OS — Ubuntu Pro (Alternative)	Ubuntu Pro with support subscription	5		
16	Kubernetes Platform	Rancher Prime (enterprise support) or OpenShift	1		
17	PostgreSQL Enterprise Support	Optional enterprise support subscription	1		
18	CI/CD Platform	GitLab Ultimate (self-hosted) or equivalent	1		
19	Any additional commercial software	Vendor to specify	—		

SOFTWARE & LICENSES TOTAL (Year 1)

8.3 AI Models & API Services (Optional)

S.No	Service	Usage	Est. Monthly Volume	Annual Price (INR)
20	Google Gemini API (Optional)	Handwriting OCR / translation fallback	10,000 docs/month	
21	Claude Sonnet API (Optional)	Legal reasoning fallback	5,000 requests/month	
22	Other API Services (Optional)	Vendor to specify	—	

8.4 Professional Services

A. Development & Integration

S.No	Item	Description	Effort (PM)	Rate (INR/PM)	Total (INR)
23	Project Management	End-to-end project coordination	3		
24	UI/UX Design	Dashboard, document viewer, annotation interface	2		
25	Frontend Development	React application with PWA capabilities	4		
26	Backend Development	FastAPI, orchestration, AI gateway	4		
27	AI Integration & Prompt Engineering	Model integration, prompt development, testing	3		
28	Model Fine-Tuning	Fine-tune Llama 3, TrOCR, NER models on domain data	2		
29	Digital Signature Integration	eSign / DSC integration	1		

S.No	Item	Description	Effort (PM)	Rate (INR/PM)	Total (INR)
30	QA & Testing	Functional, performance, security testing	3		

B. Deployment & Infrastructure Setup

S.No	Item	Description	Effort (PM)	Rate (INR/PM)	Total (INR)
31	Infrastructure Setup	Server installation, Kubernetes configuration, networking	1		
32	Security Hardening	Firewall configuration, encryption setup, access controls	1		
33	Backup & DR Setup	Backup configuration, DR testing	0.5		

C. Training & Documentation

S.No	Item	Description	Effort (PM)	Rate (INR/PM)	Total (INR)
34.	User Training	IO training workshops, user manuals, video tutorials	2		
35.	AI Admin Training	Admin console, prompt management, model updates	1		
36.	Technical Documentation	Architecture docs, API docs, runbooks	1		

PROFESSIONAL SERVICES TOTAL (Items 23–37)	
---	--

8.5 Annual Maintenance & Support (AMC) — Year 2 Onwards

S.No	Item	Description	Annual Price (INR)
37.	Hardware Warranty Extension	Server, storage, networking (Year 2 onwards)	
38.	Software Support	RHEL/Ubuntu Pro, Rancher, GitLab (as applicable)	
39.	Platform Support	Application bug fixes, security patches, minor enhancements	
40.	AI Model Updates	Model retraining, prompt optimisation, performance monitoring	
41.	Infrastructure Support	Power, cooling, facility costs (on-premise, if applicable)	

AMC TOTAL (Annual — Year 2 onwards)	
-------------------------------------	--

8.6 Optional Items

S.No	Item	Description	Priority	Price (INR)
O1	Additional GPU	Second NVIDIA RTX 6000 Ada for larger models (70B+)	Medium	
O2	High-Availability Storage	All-flash SAN with replication	Medium	
O3	LTO Tape Library	Offline archival for legal compliance	Low	
O4	Mobile Application (iOS/Android)	Field IO access app	Medium	
O5	Enhanced CCTNS Integration	Bidirectional sync with conflict resolution	High	
O6	AI Training Data Annotation	Professional annotation services for fine-tuning	Medium	
O7	Disaster Recovery Site	Secondary data centre or colocation	Low	
O8	Security Audit & Penetration Testing	Third-party security assessment	High	

8.7 Grand Summary — Vendor Quote

S.No	Category	Total Quoted (INR, excl. GST)
1	Hardware Infrastructure (Items 1–13)	
2	Software & Licenses (Items 14–19) — Year 1	
3	AI Models & API Services (Items 20–22) — Annual (if applicable)	
4	Professional Services (Items 23–37)	
5	Annual Maintenance & Support (Items 38–42) — Year 2	
TOTAL YEAR 1 (excl. GST & Optional Items)		
GST Amount (18% or as applicable)		
GRAND TOTAL YEAR 1 (incl. GST)		

9. Data Confidentiality Obligations

All test data, pilot data, and operational data accessed during any pre-contract demonstration or post-contract deployment are subject to strict confidentiality obligations under applicable law. No data shall be transmitted outside the organization's network boundary without prior written authorisation. Vendors found to have violated these obligations will be disqualified and liable for legal action.

Hyderabad City Police reserves the right to modify, cancel, or re-issue this RFQ at any time without assigning reasons. No vendor shall have any claim against HCP for costs incurred in preparation of bids. Award of contract is subject to administrative and financial approvals and emergent business requirements.

— End of Document —

Technical Consultant | Office of the Commissioner of Police | Hyderabad City Police